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# Hands-On Machine Learning with Scikit-Learn and PyTorch

Concepts, Tools, and Techniques  
to Build Intelligent Systems

Aurélien Géron

## **Praise for *Hands-On Machine Learning with Scikit-Learn and PyTorch***

*This book is an excellent starting point for beginners looking to understand the essential history and foundational concepts of machine learning. With well-structured code sections and practical examples, it takes readers from the basics to cutting-edge machine learning and deep learning techniques, leveraging PyTorch and Scikit-Learn for hands-on implementation.*

—Louis-François Bouchard, educator and cofounder and CTO at Towards AI

*Géron strikes the sweet spot: practical Scikit-Learn and PyTorch implementations that teach concepts, balanced with theory that clarifies rather than overwhelms. From first principles to state-of-the-art methods, this hands-on approach gets you productive quickly. This is the book I recommend for getting started in ML.*

—Ulf Bissbort, cofounder and CTO at ZefHub

*This book is your ultimate map for navigating the uncharted world of machine learning. Keep it within reach.*

—Haesun Park, Microsoft AI MVP, Google Cloud Champion Innovator

*This book launched a generation of ML practitioners. Brilliantly updated to cover PyTorch, it is once again the definitive hands-on guide to the field.*

—Tarun Narayanan, machine learning engineer, Amazon AGI

*A true bible for beginners in machine learning, this book not only provides clear explanations and hands-on examples but also uses thoughtfully designed figures to simplify complex concepts, making it an indispensable resource for building a strong foundation.*

—Meetu Malhotra, Harrisburg University of Science and Technology, PA USA

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**Aurélien Géron**

**Hands-On Machine Learning with Scikit-Learn and PyTorch**

by Aurélien Géron

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# Preface

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In 2006, Geoffrey Hinton et al. published [a paper<sup>1</sup>](#) showing how to train a deep neural network capable of recognizing handwritten digits with state-of-the-art precision (>98%). They branded this technique “deep learning”. A deep neural network is a (very) simplified model of our cerebral cortex, composed of a stack of layers of artificial neurons. Training a deep neural net was widely considered impossible at the time,<sup>2</sup> and most researchers had abandoned the idea in the late 1990s. This paper revived the interest of the scientific community, and before long many new papers demonstrated that deep learning was not only possible, but capable of mind-blowing achievements that no other machine learning (ML) technique could hope to match (with the help of tremendous computing power and great amounts of data). This enthusiasm soon extended to many other areas of machine learning.

A decade later, machine learning had already conquered many industries, ranking web results, recommending videos to watch and products to buy, sorting items on production lines, sometimes even driving cars. Machine learning often made the headlines, for example when DeepMind’s AlphaFold machine learning system solved a long-standing protein-folding problem that had stumped researchers for decades. But most of the time, machine learning was just working discretely in the background. However, another decade later came the rise of AI assistants: from ChatGPT in 2022, Gemini, Claude, and Grok in 2023, and many others since then. AI has now truly taken off and it is rapidly transforming every single industry: what used to be sci-fi is now very real.<sup>3</sup>

## Machine Learning in Your Projects

So, naturally you are excited about machine learning and would love to join the party! Perhaps you would like to give your homemade robot a brain of its own? Make it recognize faces? Or learn to walk around?

Or maybe your company has tons of data (user logs, financial data, production data, machine sensor data, hotline stats, HR reports, etc.), and more than likely you could unearth some hidden gems if you just knew where to look. With machine learning, you could accomplish the following **and much more**:

- Segment customers and find the best marketing strategy for each group.
- Recommend products for each client based on what similar clients bought.
- Detect which transactions are likely to be fraudulent.
- Forecast next year’s revenue.

- Predict peak workloads and suggest optimal staffing levels.
- Build a chatbot to assist your customers.

Whatever the reason, you have decided to learn machine learning and implement it in your projects. Great idea!

## Objective and Approach

This book assumes that you know close to nothing about machine learning. Its goal is to give you the concepts, tools, and intuition you need to implement programs capable of *learning from data*.

We will cover a large number of techniques, from the simplest and most commonly used (such as linear regression) to some of the deep learning techniques that regularly win competitions. For this, we will be using Python—the leading language for data science and machine learning—as well as open source and production-ready Python frameworks:

- **Scikit-Learn** is very easy to use, yet it implements many machine learning algorithms efficiently, so it makes for a great entry point to learning machine learning. It was created by David Cournapeau in 2007, then led by a team of researchers at the French Institute for Research in Computer Science and Automation (Inria), and recently Probabl.ai.
- **PyTorch** is a powerful and flexible library for deep learning. It makes it possible to train and run all sorts of neural networks efficiently, and it can distribute the computations across multiple GPUs (graphics processing units). PyTorch (PT) was developed by Facebook's AI Research lab (FAIR) and first released in 2016. It evolved from Torch, an older framework coded in Lua. In 2022, PyTorch was transitioned to the PyTorch Foundation, under the Linux Foundation, to promote community-driven development.

We will also use these open source machine learning libraries along the way:

- **XGBoost** in **Chapter 6** to implement a powerful technique called *gradient boosting*.
- **Hugging Face** libraries in **Chapters 13** and **15** to download datasets and pretrained models, including transformer models. Transformers are incredibly powerful and versatile, and they are the main building block of virtually all AI assistants today.
- **Gymnasium** in **Chapter 19** for reinforcement learning (i.e., training autonomous agents).

The book favors a hands-on approach, growing an intuitive understanding of machine learning through concrete working examples and just a little bit of theory.

TIP

While you can read this book without picking up your laptop, I highly recommend you experiment with the code examples.

## Code Examples

All the code examples in this book are open source and available online at <https://github.com/ageron/handson-mlp>, as Jupyter notebooks. These are interactive documents containing text, images, and executable code snippets (Python in our case). The easiest and quickest way to get started is to run these notebooks using Google Colab: this is a free service that allows you to run any Jupyter notebook directly online without having to install anything on your machine. All you need is a web browser and a Google account.

### NOTE

In this book, I will assume that you are using Google Colab, but I have also tested the notebooks on other online platforms such as Kaggle and Binder, so you can use those if you prefer. Alternatively, you can install the required libraries and tools (or the Docker image for this book) and run the notebooks directly on your own machine. See the instructions at <https://homl.info/install-p>.

This book is here to help you get your job done. If you wish to use additional content beyond the code examples, and that use falls outside the scope of fair use guidelines, (such as selling or distributing content from O'Reilly books, or incorporating a significant amount of material from this book into your product's documentation), please reach out to O'Reilly for permission, at [permissions@oreilly.com](mailto:permissions@oreilly.com).

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## Prerequisites

This book assumes that you have some Python programming experience. If you don't know Python yet, <https://learnpython.org> is a great place to start. The official tutorial on [Python.org](https://python.org) is also quite good.

This book also assumes that you are familiar with Python's main scientific libraries—in particular, [NumPy](#), [pandas](#), and [Matplotlib](#). If you have never used these libraries, don't worry; they're easy to learn, and I've created a tutorial for each of them. You can access them online at <https://homl.info/tutorials-p>.

Moreover, if you want to fully understand how the machine learning algorithms work (not just how to use them), then you should have at least a basic understanding of a few math concepts, especially linear algebra. Specifically, you should know what vectors and matrices are, and how to perform some simple operations like adding vectors, or transposing and multiplying matrices. If you need a quick

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introduction to linear algebra (it's really not rocket science!), I provide a tutorial at <https://homl.info/tutorials-p>. You will also find a tutorial on differential calculus, which may be helpful to understand how neural networks are trained, but it's not entirely essential to grasp the important concepts. This book also uses other mathematical concepts occasionally, such as exponentials and logarithms, a bit of probability theory, and some basic concepts from statistics, but nothing too advanced. If you need help on any of these, please check out <https://khanacademy.org>, which offers many excellent and free math courses online.

## Roadmap

This book is organized in two parts. **Part I, “The Fundamentals of Machine Learning”**, covers the following topics:

- What machine learning is, what problems it tries to solve, and the main categories and fundamental concepts of its systems
- The steps in a typical machine learning project
- Learning by fitting a model to data
- Minimizing a cost function (i.e., a measure of prediction errors)
- Handling, cleaning, and preparing data
- Selecting and engineering features (i.e., data fields)
- Selecting a model and tuning hyperparameters using cross-validation (e.g., training many model variants and choosing the one that performs best on data it didn't see during training)
- The challenges of machine learning, in particular underfitting and overfitting (the bias/variance trade-off)
- The most common learning algorithms: linear and polynomial regression, logistic regression,  $k$ -nearest neighbors, decision trees, random forests, and ensemble methods
- Reducing the dimensionality of the training data to fight the “curse of dimensionality”
- Other unsupervised learning techniques, including clustering, density estimation, and anomaly detection

**Part II, “Neural Networks and Deep Learning”**, covers the following topics:

- What neural nets are and what they're good for
- Building and training deep neural nets using PyTorch

- The most important neural net architectures: feedforward neural nets for tabular data; convolutional nets for computer vision; recurrent nets and long short-term memory (LSTM) nets for sequence processing; encoder-decoders, transformers, state space models (SSMs), and hybrid architectures for natural language processing, vision, and more; autoencoders, generative adversarial networks (GANs), and diffusion models for generative learning
- How to build an agent (e.g., a bot in a game) that can learn good strategies through trial and error, using reinforcement learning
- Loading and preprocessing large amounts of data efficiently

The first part is based mostly on Scikit-Learn; the second part uses mostly PyTorch.

### CAUTION

Don't jump into deep waters too hastily: deep learning is no doubt one of the most exciting areas in machine learning, but you should master the fundamentals first. Moreover, many problems can be solved quite well using simpler techniques such as random forests and ensemble methods (discussed in **Part I**). Deep learning is best suited for complex problems such as image recognition, speech recognition, or natural language processing, and it often requires a lot of data, computing power, and patience (unless you can leverage a pretrained neural network, as you will see).

If you are particularly interested in one topic and want to reach it as quickly as possible, **Figure P-1** will show you which chapters you must read first, and which ones you can safely skip.



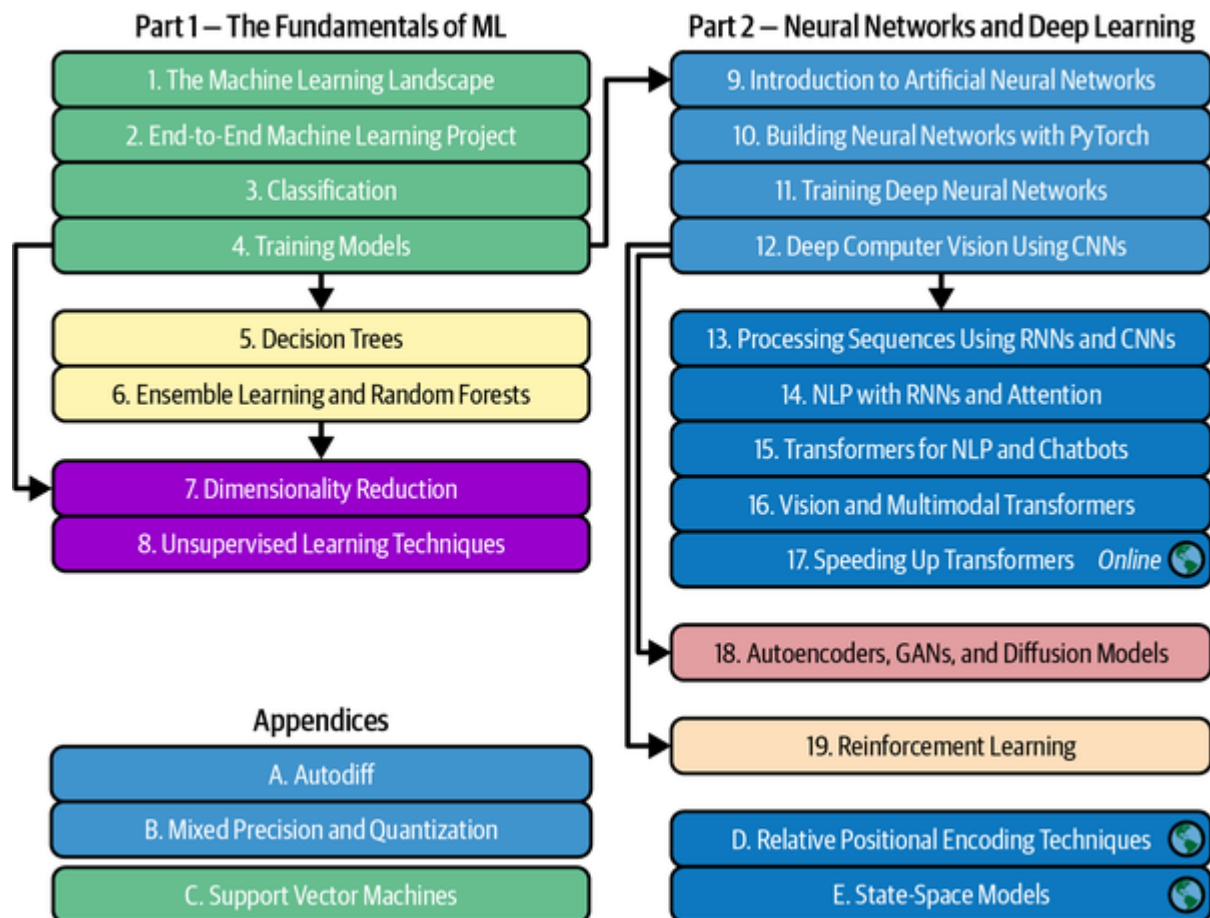


Figure P-1. Chapter dependencies

## Changes Between the TensorFlow and PyTorch Versions

I wrote three TensorFlow (TF) editions of this book, published by O'Reilly in 2017, 2019, and 2022. TF was the leading deep learning library for many years, used internally by Google and therefore optimized for production at scale. But PyTorch has gradually taken the lead, owing to its simplicity, flexibility and openness: it now dominates research papers and open source projects, which means that most new models are available in PyTorch first. As a result, the industry has also gradually shifted toward PyTorch.

In recent years, Google has reduced its investments in TensorFlow, and focused more on JAX, another excellent deep learning library with a great mix of qualities for both research and production. However, its adoption is still low compared to PyTorch.

This is why I chose to use PyTorch this time around! O'Reilly and I decided to make it the first edition of a new PyTorch series rather than the fourth edition of the original series. This leaves the door open for a JAX series or perhaps a new edition for the TF series (time will tell if either are needed).

If you have already read the latest TensorFlow version of this book, here are the main changes you will find in this book (see <https://homl.info/changes-p> for more details):

- All the code in the book was updated to recent library versions.
- All the code in **Part II** was migrated from TensorFlow and Keras to PyTorch. There were significant changes in all of these chapters.
- TensorFlow-specific content was removed, including former Chapters 12 and 13, and former Appendices C and D.
- **Chapter 10** now introduces PyTorch.
- I also added three new chapters on transformers:
  - **Chapter 15** covers transformers for natural language processing, including how to build a chatbot.
  - **Chapter 16** presents vision transformers and multimodal transformers.
  - **Chapter 17**, available online at <https://homl.info/>, discusses several advanced techniques to speed up and scale up transformers. This includes FlashAttention, mixture of experts (MoE), low-rank adaptation (LoRA), and many more.
- There are also three new appendices: **Appendix B** explains how to shrink models so they can run faster and fit on smaller devices, “Relative Positional Encoding” discusses advanced positional encoding techniques for transformers, and “State-Space Models (SSMs)” presents state-space models.
- To make room for the newer content, the chapter on support vector machines (SVMs) was **moved online** and renamed to Appendix C; the last two appendices are also online at the same URL, and the deployment chapter was partially merged into **Chapter 10**.

The three editions of the TensorFlow/Keras version of this book are nicknamed homl1, homl2, and homl3. This book, which is the first edition of the PyTorch version, is nicknamed homlp. Try saying that three times in a row.

#### NOTE

Most of the changes compared to the latest TensorFlow edition are in the second part of the book. If you have read homl3, then don’t expect big changes in the first part of the book: the fundamental concepts of machine learning haven’t changed since 2022.

## Other Resources

Many excellent resources are available to learn about machine learning. For example, Andrew Ng’s **ML course on Coursera** is amazing, although it requires a significant time investment.

There are also many interesting websites about machine learning, including Scikit-Learn's exceptional [User Guide](#). You may also enjoy [Dataquest](#), which provides very nice interactive tutorials, and countless ML blogs and YouTube channels.

There are many other introductory books about machine learning. In particular:

- Joel Grus's *Data Science from Scratch*, 2nd edition (O'Reilly), presents the fundamentals of machine learning and implements some of the main algorithms in pure Python (from scratch, as the name suggests).
- Stephen Marsland's *Machine Learning: An Algorithmic Perspective*, 2nd edition (Chapman & Hall), is a great introduction to machine learning, covering a wide range of topics in depth with code examples in Python (also from scratch, but using NumPy).
- Sebastian Raschka's *Machine Learning with PyTorch and Scikit-Learn*, 1st edition (Packt Publishing), is also a great introduction to machine learning using Scikit-Learn and PyTorch.
- François Chollet's *Deep Learning with Python*, 3rd edition (Manning), is a very practical book that covers a large range of topics in a clear and concise way, as you might expect from the author of the excellent Keras library. It favors code examples over mathematical theory.
- Andriy Burkov's *The Hundred-Page Machine Learning Book* (self-published) is very short but covers an impressive range of topics, introducing them in approachable terms without shying away from the math equations.
- Yaser S. Abu-Mostafa, Malik Magdon-Ismail, and Hsuan-Tien Lin's *Learning from Data* (AMLBook) is a more theoretical approach to ML that provides deep insights, in particular on the bias/variance trade-off (see [Chapter 4](#)).
- Stuart Russell and Peter Norvig's *Artificial Intelligence: A Modern Approach*, 4th edition (Pearson), is a great (and huge) book covering an incredible amount of topics, including machine learning. It helps put ML into perspective.
- Jeremy Howard and Sylvain Gugger's *Deep Learning for Coders with fastai and PyTorch* (O'Reilly) provides a wonderfully clear and practical introduction to deep learning using the fastai and PyTorch libraries.
- Andrew Ng's *Machine Learning Yearning* is a free ebook that provides a thoughtful exploration of machine learning, focusing on the practical considerations of building and deploying models, including data quality and long-term maintenance.
- Lewis Tunstall, Leandro von Werra, and Thomas Wolf's *Natural Language Processing with Transformers: Building Language Applications with Hugging Face* (O'Reilly) is a great practical dive into transformers using popular libraries by Hugging Face.

- Jay Alammar and Maarten Grootendorst's *Hands-On Large Language Models* is a beautifully illustrated book on LLMs, covering everything you need to know to understand, train, fine-tune, and use LLMs across a wide variety of tasks.

Finally, joining ML competition websites such as [Kaggle.com](https://www.kaggle.com) will allow you to practice your skills on real-world problems, with help and insights from some of the best ML professionals out there.

## Conventions Used in This Book

The following typographical conventions are used in this book:

### *Italic*

Indicates new terms, URLs, email addresses, filenames, and file extensions.

### `Constant width`

Used for program listings, as well as within paragraphs to refer to program elements such as variable or function names, databases, data types, environment variables, statements, and keywords.

### **Constant width bold**

Shows commands or other text that should be typed literally by the user.

### *Constant width italic*

Shows text that should be replaced with user-supplied values or by values determined by context.

### *Punctuation*

To avoid any confusion, punctuation appears outside of quotes throughout the book. My apologies to the purists.

#### **TIP**

This element signifies a tip or suggestion.

#### **NOTE**

This element signifies a general note.

#### **WARNING**

This element indicates a warning or caution.

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### NOTE

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## Acknowledgments

Never in my wildest dreams did I imagine that the three TensorFlow editions of this book would reach such a large audience. I received so many messages from readers, many asking questions, some kindly pointing out errata, and most sending me encouraging words. I cannot express how grateful I am to all these readers for their tremendous support. Thank you all so very much! Please do not hesitate to [file issues on GitHub](#) if you find errors in the code examples (or just to ask questions), or to submit [errata](#) if you find errors in the text. Some readers also shared how this book helped them get their first job, or how it helped them solve a concrete problem they were working on. I find such feedback incredibly motivating. If you find this book helpful, I would love it if you could share your story with me, either privately (e.g., via [LinkedIn](#)) or publicly (e.g., tweet me at [@aureliengeron](#) or write an [Amazon review](#)).

Huge thanks as well to all the generous people who offered their time and expertise to review this book, correcting errors and making countless suggestions. They made this book so much better: Jeremy Howard, Haesun Park, Omar Sanseviero, Lewis Tunstall, Leandro Von Werra, and Sam Witteveen reviewed the table of contents and helped me refine the scope of the book. Hesam Hassani, Ashu Jha, Meetu Malhotra, and Ammar Mohanna reviewed the first part, while Ulf Bissbort, Louis-Francois Bouchard, Luba Elliot, Thomas Lacombe, Tarun Narayanan, Marco Tabor, and my dear brother Sylvain reviewed the second part. Special thanks to Haesun Park, who reviewed every single chapter. You are all amazing!

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1 Geoffrey E. Hinton et al., “A Fast Learning Algorithm for Deep Belief Nets”, *Neural Computation* 18 (2006): 1527–1554.

2 Despite the fact that Yann LeCun's deep convolutional neural networks had worked well for image recognition since the 1990s, although they were not as general-purpose.

3 Geoffrey Hinton was awarded the 2018 Turing Award (with Yann LeCun and Yoshua Bengio) and the 2024 Nobel Prize in Physics (with John Hopfield) for early work on neural networks back in the 1980s. DeepMind's founder and CEO Demis Hassabis and director John Jumper were awarded the 2024 Nobel



Prize in Chemistry for their work on AlphaFold. They shared this Nobel Prize with another protein researcher, David Baker.

# Part I. The Fundamentals of Machine Learning

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## Chapter 1. The Machine Learning Landscape

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Not so long ago, if you had picked up your phone and asked it to tell you the way home, it would have ignored you—and people would have questioned your sanity. But machine learning is no longer science fiction: billions of people use it every day. And the truth is it has actually been around for decades in some specialized applications, such as optical character recognition (OCR). The first ML application that really became mainstream, improving the lives of hundreds of millions of people, discretely took over the world back in the 1990s: the *spam filter*. It's not exactly a self-aware robot, but it does technically qualify as machine learning: it has actually learned so well that you seldom need to flag an email as spam anymore. Then thanks to big data, hardware improvements, and a few algorithmic innovations, hundreds of ML applications followed and now quietly power hundreds of products and features that you use regularly: voice prompts, automatic translation, image search, product recommendations, and many more. And finally came ChatGPT, Gemini (formerly Bard), Claude, Perplexity, and many other chatbots: AI is no longer just powering services in the background, it *is* the service itself.

Where does machine learning start and where does it end? What exactly does it mean for a machine to *learn* something? If I download a copy of all Wikipedia articles, has my computer really learned something? Is it suddenly smarter? In this chapter I will start by clarifying what machine learning is and why you may want to use it.

Then, before we set out to explore the machine learning continent, we will take a look at the map and learn about the main regions and the most notable landmarks: supervised versus unsupervised learning and their variants, online versus batch learning, instance-based versus model-based learning. Then we will look at the workflow of a typical ML project, discuss the main challenges you may face, and cover how to evaluate and fine-tune a machine learning system.

This chapter introduces a lot of fundamental concepts (and jargon) that every data scientist should know by heart. It will be a high-level overview (it's the only chapter without much code), all rather

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simple, but my goal is to ensure everything is crystal clear to you before we continue on to the rest of the book. So grab a coffee and let's get started!

#### TIP

If you are already familiar with machine learning basics, you may want to skip directly to **Chapter 2**. If you are not sure, try to answer all the questions listed at the end of the chapter before moving on.

## What Is Machine Learning?

Machine learning is the science (and art) of programming computers so they can *learn from data*.

Here is a slightly more general definition:

*[Machine learning is the] field of study that gives computers the ability to learn without being explicitly programmed.*

—Arthur Samuel, 1959

And a more engineering-oriented one:

*A computer program is said to learn from experience  $E$  with respect to some task  $T$  and some performance measure  $P$ , if its performance on  $T$ , as measured by  $P$ , improves with experience  $E$ .*

—Tom Mitchell, 1997

Your spam filter is a machine learning program that, given examples of spam emails (flagged by users) and examples of regular emails (nospam, also called “ham”), can learn to flag spam. The examples that the system uses to learn are called the *training set*. Each training example is called a *training instance* (or *sample*). The part of a machine learning system that learns and makes predictions is called a *model*. Neural networks and random forests are examples of models.

In this case, the task  $T$  is to flag spam for new emails, the experience  $E$  is the *training data*, and the performance measure  $P$  needs to be defined; for example, you can use the ratio of correctly classified emails. This particular performance measure is called *accuracy*, and it is often used in classification tasks (we will discuss several others in **Chapter 3**).

If you just download a copy of all Wikipedia articles, your computer has a lot more data, but it is not suddenly better at any task. This is not machine learning.

## Why Use Machine Learning?

Consider how you would write a spam filter using traditional programming techniques (**Figure 1-1**):  
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